F79MA Coursework: Assessed Project 2

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1. Introduction

In many areas of biology, scientists are interested in the number of times a specific genetic mutation occurs within a fixed length of an RNA sequence. Due to the fact that these outcomes are non-negative integers representing event counts, it is natural to consider probabilistic models designed for count data. A simple and widely used model is the Poisson distribution, which assumes that events occur independently and at a constant average rate. However, real biological data often requires consideration of a spectrum, with each count not necessarily being independently and identically distributed. In such cases, the Poisson model becomes inadequate because it cannot capture the extra variability present in the observations.

A more flexible alternative is the negative binomial distribution, which introduces variability in the underlying rate of occurrence. One way to justify the negative binomial model is by considering the Poisson rate parameter is itself assumed to follow a Gamma distribution. This Poisson–Gamma mixture naturally leads to a negative binomial marginal distribution for the observed counts and allows us to model both the mean behaviour and the variation between sequences more realistically.

The aim of this project is to perform a bayesian analysis of the provided count data for genetic mutations. Specifically, we treat the data as independent and identically distributed observations from a negative binomial distribution with a fixed parameter $k = 5$ and an unknown success probability p. The analysis involves deriving the Jeffreys' prior for p, obtaining the posterior distribution given the observed data, and evaluating the resulting posterior predictive distribution for future observations. Graphical comparisons between prior, posterior, and predictive distributions will be used to assess model behaviour and adequacy.

2. Mathematical Derivations

In Section 2.1, we explore why the Gamma-Poisson mixture would be more appropriate to model count data in various contexts rather than directly using a Poisson distribution. In Section 2.2, we derive the Jeffreys' prior and comment on the subsequent results that follow. In Section 2.3, we derive the posterior density for parameter p using Jeffreys' Prior. In Section 2.4, we derive the posterior predictive distribution for an unseen count z.

2.1 Negative Binomial Distribution Posterior-Predictive Mass Function

To explain in the language of bayesian statistics, let $\underline{X}$ be a random sample of count data from $\prod(\lambda)$, we choose the prior $\lambda \sim \Gamma(\alpha, \beta)$ as the rate of the Poisson distribution. Then, given a random sample $\underline{x}$ we have,

$$\begin{aligned}
\pi(\lambda|\underline{x}) &\propto \pi(\lambda)L(\lambda;\underline{x}) \\
&\propto \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}\frac{1}{\prod x_i!}\lambda^{\sum x_i}e^{-n\lambda}, \quad \text{where } n \text{ is the sample size,} \\
&\propto \lambda^{\alpha+\sum x_i-1}e^{-(\beta+n)\lambda},
\end{aligned}$$

implying the posterior distribution is $\lambda|\underline{x} \sim \Gamma(\alpha', \beta')$, where $\alpha' = \alpha + \sum x_i$ and $\beta' = \beta + n$.

Therefore, the posterior-predictive mass function for a new observation of count data $Y \sim \prod(\lambda)$ is,

$$\begin{aligned} f_Y(y|\underline{x}) = P(Y = y|\underline{x}) &= \int_0^\infty P(y = y|\lambda)\pi(\lambda|\underline{x})\ d\lambda \\ &= \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!} \cdot \frac{\beta'^{\alpha'}}{\Gamma(\alpha')}\lambda^{\alpha'-1}e^{-\beta'\lambda}\ d\lambda \\ &= \frac{\beta'^{\alpha'}}{\Gamma(\alpha')y!}\int_0^\infty \lambda^{\alpha'+y-1}e{-}(\beta'+1)\lambda\ d\lambda \\ &= \frac{\beta'^{\alpha'}}{\Gamma(\alpha')y!} \cdot \frac{\Gamma(\alpha'+y)}{(\beta'+1)^{\alpha'+y}} \\ &= \frac{\Gamma(\alpha'+y)}{\Gamma(\alpha')y!}\left(\frac{\beta'}{\beta'+1}\right)^{\alpha'}\left(\frac{1}{\beta'+1}\right)^{y} \end{aligned}$$

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showing it can be modelled by $f_Y(y|\underline{x}) \sim NegBin(k, p)$ with $k = \alpha'$ and $p = \frac{\beta'}{\beta'+1}$.

This may be more appropriate than directly using a Poisson distribution because Poisson distribution assumes counts vary around the mean with equal variance, with all observations using the same rate (λ), while modelling using the Gamma–Poisson mixture (which follows the negative binomial distribution) allows us to capture a range of possible rates (λ) for each observation, which is more realistic in a real-life context.

2.2 Jeffreys' Prior

We attempt to find the Jeffreys' Prior for the parameter p for a sample size $n = 1000$ from a negative binomial distribution with other parameter $k = 5$. We will see that variables n and k are irrelevant constants that will not affect the derivation of the Jeffreys' prior.
First, consider deriving the Fisher's information with $X \sim NegBin(k, p)$,

$$\begin{aligned} L(p;\underline{x}) &\propto \left(\frac{p}{1-p}^{nk}\right)(1-p)^{\sum x_i} \\ \implies l(p;\underline{x}) &\propto nk(ln(p) - ln(1-p)) + \sum x_i ln(1-p) \\ \implies \frac{dl}{dp} &= nk\left(\frac{1}{p} + \frac{1}{1-p}\right) - \frac{\sum x_i}{1-p} \\ \implies \frac{d^2l}{dp^2} &= nk\left(-\frac{1}{p^2} + \frac{1}{(1-p)^2}\right) - \frac{\sum x_i}{(1-p)^2} \end{aligned}$$

Since $I(p) = \mathbb{E}[-\frac{d^2l}{dp^2}]$, then,

$$\begin{aligned} I(p) &= nk\left(\frac{1}{p^2} - \frac{1}{(1-p)^2}\right) + \frac{\frac{nk}{p}}{(1-p)^2} \\ &= nk\left(\frac{1}{p^2} - \frac{1}{(1-p)^2} + \frac{1}{p(1-p)^2}\right) \\ &\propto \frac{1}{p^2(1-p)} \\ \implies \pi_J(p) &= p^{-1}(1-p)^{-\frac{1}{2}} \quad \left(\because \pi_J(p) = \sqrt{I(p)}\right) \end{aligned}$$

Comments on the resulting prior distribution: Notice that the Jeffreys' prior resembles (but is not) the $Beta(0, \frac{1}{2})$ distribution. However, since both parameters of the distribution must be positive, and $\Gamma(0) = \infty$, this proves that the Jeffreys' prior is improper.

2.3 Posterior Density for Parameter p

Using the Jeffreys' Prior derived in 2.2, the posterior density for parameter p follows a simple derivation as such,

$$\begin{aligned} \pi(p|x) &\propto \pi_J(p)L(p;\underline{x}) \\ &\propto p^{nk-1}(1-p)^{\sum x_i - nk - \frac{1}{2}} \end{aligned}$$

Therefore, $p|x \sim Beta(nk, \sum x_i - nk + \frac{1}{2})$.

2.4 Posterior Predictive Distribution for an Unseen Count z

The posterior predictive distribution $\pi(z|\underline{x})$ for an unseen count z follows as such,

$$\begin{aligned} \pi(z|\underline{x}) &= \int_0^1 \pi(p|\underline{x})P(Z=z|p)\ dp \\ &= \int_0^1 \frac{\Gamma(nk + \sum x_i - nk + \frac{1}{2})}{\Gamma(nk)\Gamma(\sum x_i - nk + \frac{1}{2})} p^{nk-1}(1-p)^{\sum x_i - nk + \frac{1}{2} - 1} \cdot \binom{z-1}{k-1} p^k (1-p)^{z-k}\ dp \\ &= \frac{\Gamma(\sum x_i + \frac{1}{2})}{\Gamma(nk)\Gamma(\sum x_i - nk + \frac{1}{2})} \cdot \binom{z-1}{k-1} \int_0^1 p^{nk+k-1}(1-p)^{\sum x_i - nk + z - k - \frac{1}{2}}\ dp \\ &= \frac{\Gamma(\sum x_i + \frac{1}{2})}{\Gamma(nk)\Gamma(\sum x_i - nk + \frac{1}{2})} \cdot \binom{z-1}{k-1} \cdot \frac{\Gamma(nk+k)\Gamma(\sum x_i - nk + z - k + \frac{1}{2})}{\Gamma(nk + k + \sum x_i - nk + z - k + \frac{1}{2})} \\ &= \binom{z-1}{k-1} \cdot \frac{\Gamma(nk+k)\Gamma(\sum x_i - nk + z - k + \frac{1}{2})\Gamma(\sum x_i + \frac{1}{2})}{\Gamma(nk)\Gamma(\sum x_i - nk + \frac{1}{2})\Gamma(\sum x_i + z + \frac{1}{2})} \\ &= \binom{z-1}{k-1} \cdot \frac{B(nk+k, \sum x_i - nk + \frac{1}{2} + z - k)}{B(nk, \sum x_i - nk + \frac{1}{2})} \quad \text{where } B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} \end{aligned}$$

To accommodate R's computing power that does not allow for extremely small values, we modify the posterior predictive distribution as such:

$$\pi(z|\underline{x}) = \binom{z-1}{k-1} \cdot \frac{B(nk+k, \sum x_i - nk + \frac{1}{2} + z - k)}{B(nk, \sum x_i - nk + \frac{1}{2})}$$
$$= \binom{z-1}{k-1} \cdot \text{Exp}\left\{ log[B(nk+k, \sum x_i - nk + \frac{1}{2} + z - k)] - log[B(nk, \sum x_i - nk + \frac{1}{2})] \right\}$$

By substituting the value of z, this function will return the probability of that specific event count happening.

3. Simulations in R

To complement this report, we have coded an R script `F79MA_Assignment2.R`. Using the seed 6014, we select a sample from the count data called `My_Data`, which we will use for our bayesian analysis.

3.1 The Maximum Likelihood Estimate (MLE) $\hat{p}$ for the Parameter p

It is given that $\hat{p} = \frac{k}{\bar{x}}$ where $\bar{x}$ is the sample mean. Therefore, substituting relevant values, with $k = 5$ gives 0.2194715 (`Quantity1`).

3.2 Calculation of the Posterior Mean

Using result from 2.3 above, $p|x \sim Beta(nk, \sum x_i - nk + \frac{1}{2})$. Therefore, the posterior mean follows as $\frac{nk}{nk+\sum x_i - nk + \frac{1}{2}} = \frac{nk}{\sum x_i + \frac{1}{2}}$. By substituting the relevant values, we get 0.2194667 which forms `Quantity2`.

Notice that the posterior mean is quite close to the MLE for parameter p. This showcases alignment between bayesian inference and the frequentist approach.

3.3 Densities of Prior and Posterior Distributions

Due to the Jeffreys' prior being an improper distribution, despite the posterior density being proper, observing the densities of both prior and posterior distributions using sample generation has proved unfruitful. Therefore, using the prior from 2.2 and the data from `My_Data`, a line graph of both densities (Figure 1), with the prior distribution as a function, and the posterior distribution calling the `dbeta` function, have been plotted to compare and contrast the shapes of prior and posterior densities.

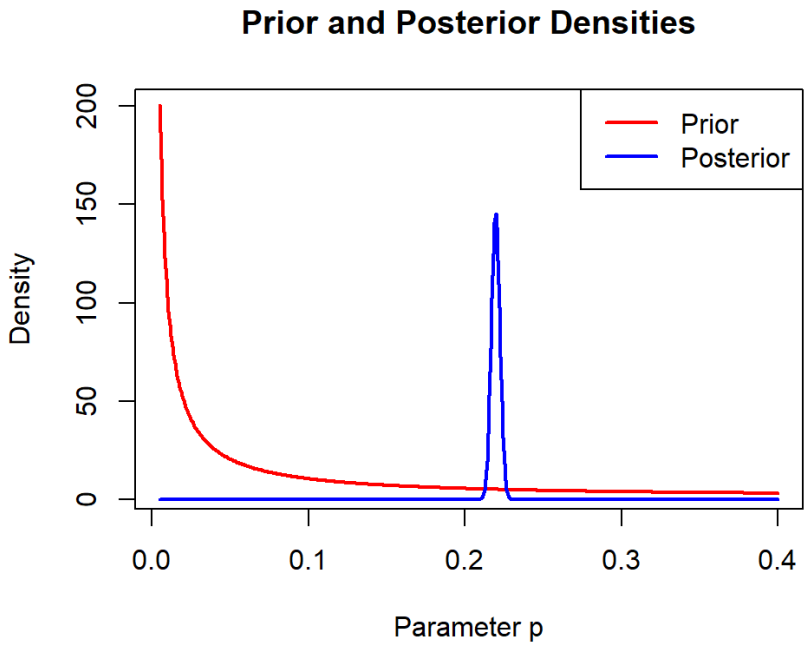


Figure 1: Prior and posterior densities for parameter p.

First, we note that the prior density is heavily weighted near $p = 0$, it does not represent real prior knowledge about the process and is improper on its own. After observing the data, the posterior is a proper Beta distribution that is tightly concentrated around the data-driven estimate of p. We can therefore conclude that the data overwhelm the vague prior and produce a precise estimate of the model parameter p, so our inference is driven by the observed counts rather than by subjective prior assumptions. This fulfills the objective as we intentionally chose to use the Jeffreys' Prior due to it being invariant to changes of parametrisation.

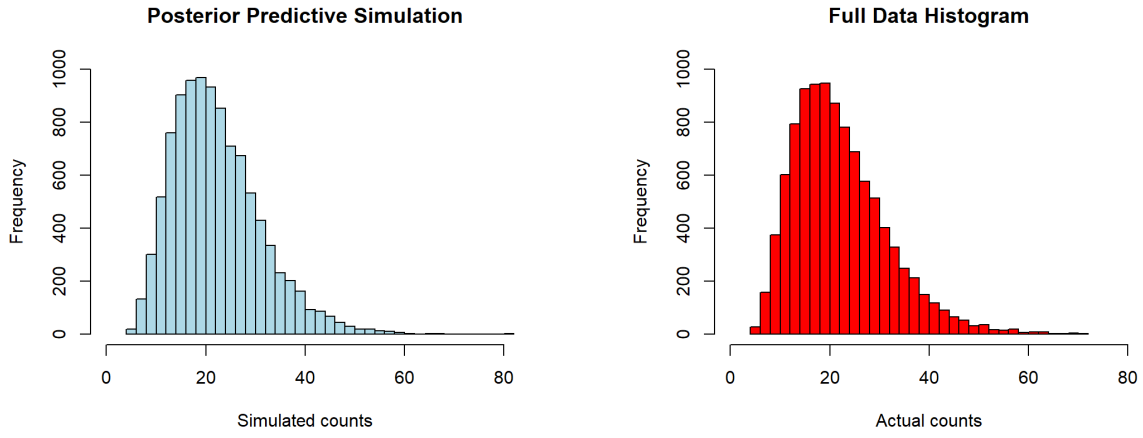
3.4 The Posterior Predictive Distribution

Using result from 2.4 above, we substitute the required values while considering z = 5, to obtain the value of $\pi(Z = 5|\underline{x}) = 0.0005099421$, which is stored as `Quantity3`. This results shows that the predictive probability that the number of counts $Z = 5$ would be 0.0005099421.

3.5 Realisations of the Posterior Predictive Distribution

After simulating 10000 realisations from the predictive distribution from 2.4, the results of the simulation and actual data are plotted as a histogram in Figure 2 as such. Due to the x variable for the negative binomial distribution in R being defined to start from 0 while our definition of the negative binomial distribution starts from k where $k = 5$ in this scenario, we have manually shifted the samples by adding 5 to all samples generated with `rnbinom` to account for this modification.

The posterior predictive distribution generated from the bayesian model provides an estimate of how well the fitted negative binomial process reproduces the observed variation in the count data. The simulated histogram of 10000 drawn from the predictive distribution closely mirrors the shape of the histogram of the full data, displaying a similar centre, spread, and overall right-skewed



(a) Simulated Results (b) Actual Data

Figure 2: Comparison between simulated results using posterior predictive distribution and actual data

form. This indicates that the model captures both the central tendency and the degree of variance present in the observed counts.

Minor discrepancies between the simulated and observed distributions are expected, particularly in the extreme tails where data are sparse. In this case, the predictive distribution slightly smooths the extreme high-count region. Nevertheless, the predictive mean and variance align well with those of the empirical data, suggesting that the fitted model is well calibrated and capable of generating realistic future observations.

4. Conclusion

This project has applied a bayesian inference framework to model count data representing the number of genetic mutations observed in RNA sequences. By assuming a negative binomial distribution with a fixed parameter $k = 5$ and an unknown success probability p, we were able to account for variability in the data, a feature not captured by the simpler Poisson model.

Using the Jeffreys' prior, which is objective and invariant to re-parameterisation, we derived a posterior distribution for p in the form of $Beta(nk, \sum x_i - nk + \frac{1}{2})$. The resulting posterior density was highly concentrated around its mean, indicating that the dataset provided strong information about the true value of p. The posterior mean (stored as `Quantity2`) was found to be close to the maximum likelihood estimate (stored as `Quantity1`), confirming consistency between bayesian and frequentist approaches in this large-sample context.

A comparison of the prior and posterior densities demonstrated the influence of the observed data: the broad and improper Jeffreys' prior was quickly dominated by the likelihood, producing a well-defined, sharply peaked posterior. The derived posterior predictive distribution successfully replicated the general shape and spread of the empirical data (as seen in the histogram comparison), suggesting that the negative binomial model provides a reasonable fit and captures the observed variability.

Overall, this analysis illustrates the usefulness of bayesian inference methods for modelling

biological count data, particularly when the data demands concern over data variability. The combination of an invariant, principled prior, the posterior, and predictive simulation further closes the gap between bayesian inference methods and the traditional frequentist approach.